

MadhanRaj D

Generative AI Engineer (RAG & Agentic Systems)

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madhan-raj-22032b27b/) | [GitHub](https://github.com/MADHANRAJ24) | [Portfolio](https://madhanraj24.github.io/my-portfolio/) | [Kaggle]

PROFESSIONAL SUMMARY

Innovative Generative AI Engineer specializing in Agentic RAG, NLP, and Multi-Agent Orchestration. Proven track record in building complex autonomous systems, including scientific research assistants and high-frequency trading bots using LangGraph. Expert in designing self-correcting AI workflows and dynamic knowledge retrieval systems. Adept at transforming complex datasets into production-ready AI insights using PyTorch, FAISS, and LLMs.

TECHNICAL SKILLS

Generative AI & LLMs: LangGraph, CrewAI, LangChain, Agentic RAG, Multi-Agent Workflows, Stateful Reasoning, Prompt Engineering, OpenAI/Groq API.

Data Science & ML: Python (Advanced), PyTorch, TensorFlow, Scikit-learn, FAISS, Sentence Transformers, NLP, Text Mining, Applied Statistics.

Tools & Frameworks: FastAPI, React, SQL (PostgreSQL/Supabase), Git, Docker, Power BI, Hadoop (Familiarity).

Languages: Python (Primary), SQL.

PROFESSIONAL EXPERIENCE

Codec Technologies (AICTE Portal) | *Data Science Intern* | Madurai, India

Oct 2025 – Present

Gained hands-on experience through the AICTE National Internship Portal, contributing to innovative data projects.

Collaborating with industry experts to build and evaluate ML models (Regression, Classification) for targeted business challenges.

Applying advanced data preprocessing and feature engineering on structured and unstructured real-world datasets.

CodeAlpha | *Data Science Virtual Intern (Completed)* | Madurai, India

Sept 2025 – Dec 2025

Successfully completed a 3-month intensive virtual internship focusing on real-world data science applications.

Strengthened core skills in data analysis, model optimization, and predictive modeling using Python and advanced ML frameworks.

Architected end-to-end machine learning pipelines and developed predictive models achieving significant improvements in business use cases.

Sfoenix Technologies | *Data Science Intern* | Remote

Executed advanced data preprocessing and feature engineering on complex datasets.

Built and evaluated ML models for specific business use cases, improving data-driven decision-making.

KEY PROJECTS

AI Scientific Assistant (Agentic RAG & Scientific Logic)

LangGraph, FAISS, Sentence Transformers, Streamlit

Engineered an advanced Agentic RAG system for scientific instrument recommendation using LangGraph for stateful, multi-stage reasoning.

Implemented Dynamic Knowledge Injection with an auto-indexing pipeline (FAISS) that hot-loads new documentation/instrument data without application restarts.

Developed a History-Aware Reasoning node that handles complex, multi-turn research dialogues and suggests complementary instrument combinations for multi-layered samples.

Integrated rule-based validation guardrails to ensure physical constraint compliance (e.g., sample state vs. vacuum compatibility), significantly improving recommendation reliability.

Super Intelligent Trading Bot (Generative AI & Finance)

LangGraph, Groq, Angel One API, FastAPI

Developed a fully autonomous multi-actor trading agent utilizing LangGraph for stateful decision-making.

Implemented a self-improving reinforcement learning loop that adjusts trading strategies based on historical performance.

Integrated "Chart Vision" for multimodal analysis of candlestick patterns and real-time sentiment analysis for trade execution.

The Swarm: LLM Powered Ecosystem (Agentic Workflows)

LangChain, OpenAI, React

Orchestrated a suite of 20+ specialized AI agents including YouTube Summarizers, Medical Chatbots, and Resume Q&A tools.

Designed complex multi-agent workflows to automate backend processes, showcasing expertise in prompt engineering and RAG.

The Sentinel: Warranty Claim Fraud Prediction (Business Process)

Scikit-learn, Pandas, Data Visualization

Built a predictive model to identify fraudulent warranty claims by analyzing regional data, product categories, and claim history.

Automated data collection and preprocessing of massive datasets to discover hidden patterns in business processes.

The Architect: Mini LLM from Scratch

PyTorch, Transformers

Engineered a Transformer-based model from the ground up, implementing custom attention mechanisms and tokenization.

EDUCATION

B.Tech in Artificial Intelligence & Data Science

Madurai, India | 80% Score

CERTIFICATIONS & ACHIEVEMENTS

Python Certified (Advanced)

Kaggle Dataset Contributor (Top % in specific categories)

Coursera Specializations (x3): Deep Learning, AI for Medicine, Applied Data Science.

HackerRank: Problem Solving (Gold Badge).

VOLUNTEER & LEADERSHIP

Contributor to open-source agentic frameworks on GitHub.

Active member of local Data Science and AI development communities.